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A Motion to Change Venue for the Riemann Hypothesis

RE-CENTERING ARITHMETIC · A MOTION WITH EXHIBITS

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$$K(m,n) = \frac{\gcd(m,n)}{\sqrt{mn}}$$

FIG. 0 THE INSTRUMENT. ONE KERNEL, ENTERED THREE TIMES: AS OVERLAP, AS SPECTRUM, AS RESPONSE.

The motion argues and proves nothing.
The exhibits prove and argue nothing.

The guiding slogan:
divisibility has a finite-place spectral shadow.

A filing that arrives with its record attached. What is proved is entered as Exhibits A through F. What is known is entered as Exhibits G and H. What remains open is entered, fenced and named, as Exhibit I.

DECLARED NOT CONTAINED

The Riemann Hypothesis is **not proved** here. No Hilbert–Pólya operator is constructed. The Weil explicit formula is not derived from the gcd kernel, and the zeta zeros are not shown to be the spectrum of a self-adjoint operator.

What the record contains, precisely: the normalized gcd kernel gives a genuine finite-place Hilbert geometry of divisibility — the Gram kernel of normalized divisibility sets in $\widehat{\mathbb{Z}}$, and the Fourier transform of an explicit product of local Poisson measures on the prime torus. Under the log-scale flow the direct finite Euler shadows fail as length-side cargo but survive exactly as matrix coefficients of a diagonal scale flow. Their logarithmic radial derivatives isolate the prime-power distribution of the explicit formula. The far-bank condition is completed Weil positivity — RH-strength, by Weil’s criterion.

THE THESIS UNDERNEATH

Cancellation does not know its sign until it is represented as length. The positivity RH demands becomes visible only in a Hilbert geometry, where positivity is inherited rather than verified. Sharpened: the operator is not the source of the arithmetic. The scale generator is nearly empty. **The state carries the information**, and at the critical boundary the state is what must be renormalized.

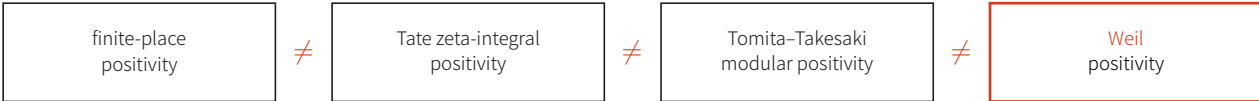


FIG. 1 FOUR POSITIVITIES. NEIGHBORING ROOMS, NOT ONE ROOM. THE MOTION SHOWS THE DOORS; THE EXHIBITS NAME THE LOCKED ONE — EXHIBIT I.

Re-centering is how arithmetic has often developed. Zero was installed, not found. Negative quantities were called absurd and used anyway. The diagonal of the unit square admitted no ratio, so a continuum was built until it did not need one. When an absence is structured, persistent, and resistant to the standing coordinates, the historical remedy has been to rebuild the coordinates around it.

■ THE NO-LEAK CONDITION

THE CLASSICAL STATEMENT, RE-EXPRESSED IN THE WORKING COORDINATE

Write a zero $\rho = \beta + i\gamma$ and follow it into logarithmic scale, with $x = e^t$ and square-root normalization:

$$\frac{x^\rho}{x^{1/2} \rho} = \frac{\overbrace{e^{(\beta-1/2)t}}^{\text{the leak}} \cdot \underbrace{e^{i\gamma t}}_{\text{oscillation}}}{\rho}.$$

If $\beta = 1/2$: pure oscillation. If $\beta \neq 1/2$: the pair contains an exponentially growing mode.

■ NO-LEAK CONDITION · WHY THE RENAMING PAYS

After square-root normalization, every zero-pair contributes only pure oscillatory modes. Not an independent criterion — the classical zero-location condition in the explicit formula’s own coordinate.
A misplaced point sounds small. A leaking mode changes the type of the system. Legibility is what the renaming buys.

■ THE ORDER OF THE RECORD

NO NEW INSTRUMENT. THE EXISTING ONES, ROUTED.

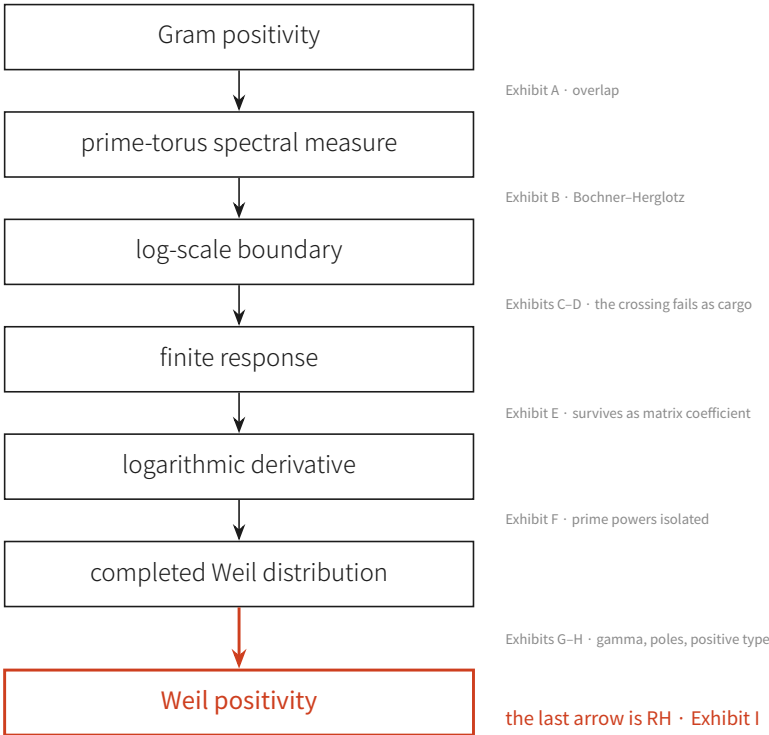


FIG. 2 EVERY ARROW BEFORE THE LAST CAN BE STATED PRECISELY.

PART II · INDEX OF EXHIBITS

EACH CLAIM ASSIGNED ITS EXHIBIT AND ITS STANDING

The reader should not have to guess which kind of sentence is being spoken. Proved mathematics, known theorems, and speculative interpretation are different currencies.

EX.	CLAIM	STATUS
A	$K(m, n) = \gcd(m, n) / \sqrt{mn}$ is positive definite	■ PROVED
B	K extends to positive-definite κ on $\mathbb{Q}_+^\times$, with explicit spectral measure on the prime torus $\mathbb{T}^{\mathcal{P}}$	■ PROVED
C	The product measures μ_σ change type at $\sigma = 1/2$; the scale flow is expelled at the same threshold	■ PROVED
D	Finite Euler shadows are positive on log-scale ...and converge directly to the completed Weil functional	■ PROVED × FALSE AS STATED
E	Finite shadows are exact matrix coefficients of a trivial scaling generator; the normalized response is the zeta distribution's symmetrized autocorrelation	■ PROVED
F	Logarithmic radial derivatives isolate the prime-power distribution of the explicit formula	■ PROVED
G	Tate: the archimedean Gaussian supplies the gamma factor; completion adds gamma, pole, trivial-zero terms	■ KNOWN
H	Weil positivity of the completed functional $\Leftrightarrow$ RH	■ KNOWN
I	A positive trace identity realizes the Weil functional; a modular flow has spectrum given by the zeta zeros	■ OPEN

READING ORDER OF THE RECORD

The finite-place spectral theorem is the unconditional core. The boundary exhibits explain why the most tempting direct limit fails, why the same object survives as response, and why the logarithmic derivative — not the raw product — is the operation that reaches the explicit formula. The RH-strength claim stays fenced in Exhibit I.

INSTRUMENTS, IN ORDER OF ENTRY:
Smith 1876 (gcd determinant) · Herglotz/Bochner (spectral measures) · Khinchine (zeta distribution) · Kakutani 1948 (product-measure dichotomy) · Tate 1950 (completion) · Weil 1952 (positivity criterion) · Tomita–Takesaki ~1970 (modular time) · Bost–Connes 1995 · Connes 1999 (trace formula on adèle classes).

■ EXHIBIT A · THE FINITE-PLACE GRAM KERNEL

DIVISIBILITY AS OVERLAP

In $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$ with Haar measure, the set $D_n = n\widehat{\mathbb{Z}}$ has measure $1/n$. Normalize: $e_n = \sqrt{n} \mathbf{1}_{D_n}$. Since $D_m \cap D_n = \text{lcm}(m, n)\widehat{\mathbb{Z}}$,

$$\langle e_m, e_n \rangle = \frac{\sqrt{mn}}{\text{lcm}(m, n)} = \frac{\gcd(m, n)}{\sqrt{mn}}, \quad \sum_{m, n} \overline{c_m} c_n K(m, n) = \left\| \sum_n c_n e_n \right\|^2 \geq 0.$$

— NOTICE THE EXPONENT · SAFETY RAIL 1

The vector $\mathbf{1}_{D_n}$ has norm $n^{-1/2}$; normalization multiplies by $\sqrt{n}$. The same half-density that later appears at the critical line already appears in the most elementary finite-place geometry. This does not prove RH — it explains why the kernel is naturally centered at exponent $1/2$. And the kernel represents overlap, **not yet the Weil functional**.

■ EXHIBIT B · THE FINITE-PLACE SPECTRAL THEOREM

DIVISIBILITY AS SPECTRUM

By unique factorization $\mathbb{Q}_+^\times \cong \bigoplus_p \mathbb{Z}$, extend the kernel to

$$\kappa(q) = \prod_p p^{-|v_p(q)|/2}, \quad \kappa(m/n) = \frac{\gcd(m, n)}{\sqrt{mn}}.$$

On the dual — the infinite prime torus $\widehat{G} \cong \mathbb{T}^{\mathcal{P}}$ — put one Poisson measure per prime, at radius $a = p^{-1/2}$:

$$d\mu_p(e^{i\theta}) = \frac{1 - p^{-1}}{|1 - p^{-1/2}e^{i\theta}|^2} \frac{d\theta}{2\pi}, \quad \int_{\mathbb{T}} z^k d\mu_p(z) = p^{-|k|/2}.$$

$$\text{Finite-place divisibility spectrum.} \quad \kappa(q) = \int_{\mathbb{T}^{\mathcal{P}}} \chi_z(q) d\mu(z), \quad \mu = \bigotimes_p \mu_p.$$

Two exact models of one kernel:

$$\frac{\gcd(m, n)}{\sqrt{mn}} = \underbrace{\langle \sqrt{m} \mathbf{1}_{m\widehat{\mathbb{Z}}}, \sqrt{n} \mathbf{1}_{n\widehat{\mathbb{Z}}} \rangle}_{\text{overlap}} = \underbrace{\int_{\mathbb{T}^{\mathcal{P}}} \chi_m \overline{\chi_n} d\mu}_{\text{spectrum}}.$$

FIG. 3 THE UNCONDITIONAL CORE. EVERYTHING DOWNSTREAM ROUTES THROUGH THIS IDENTITY.

■ EXHIBIT C · THE CRITICAL BOUNDARY

HAAR SINGULARITY AND THE EXPELLED SCALE FLOW

Vary the radius: $\mu_\sigma = \bigotimes_p P_{p^{-\sigma}} d\theta_p / 2\pi$. Kakutani's criterion reduces everything to one series:

$$H_p(\sigma) = 1 - \frac{1}{4}p^{-2\sigma} + O(p^{-4\sigma}), \quad \prod_p H_p(\sigma) > 0 \iff \sum_p p^{-2\sigma} < \infty.$$

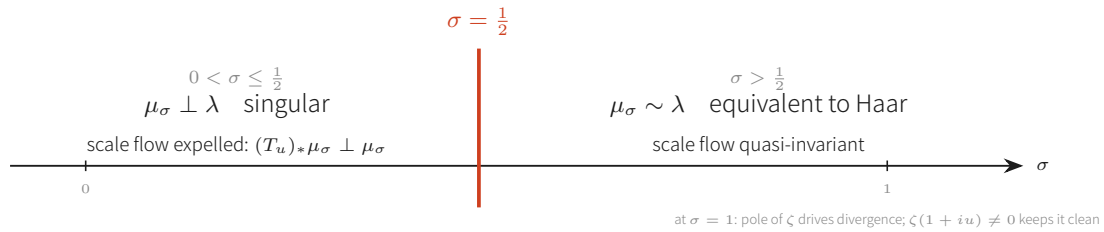


FIG. 4 A MEASURE-THEORETIC PHASE BOUNDARY AT THE CRITICAL EXPONENT.

The sharper statement compares μ_σ with its own translate under the log-scale shift $T_u: (z_p)_p \mapsto (p^{iu} z_p)_p$. Kakutani reduces quasi-invariance to the convergence of

$$\sum_p p^{-2\sigma} (1 - \cos(u \log p)),$$

which converges for $\sigma > \frac{1}{2}$ and diverges for every $u \neq 0$ at and below the threshold. At $\sigma = \frac{1}{2}$ the divergent half is powered by the pole of ζ at $s = 1$, and kept clean by the Hadamard–de la Vallée Poussin nonvanishing $\zeta(1 + iu) \neq 0$.

■ PAUSE · WHAT THIS DOES AND DOES NOT SAY

The critical normalization $p^{-1/2}$ is the exact product-measure boundary. **The critical line is not imported as metaphor. It appears as a measure-theoretic phase boundary.** At the critical exponent the measure is not merely singular relative to Haar — the scale flow itself refuses to act within its measure class. The vacuum is not transported across the boundary as a measure. This does not prove RH; it locates the difficulty.

■ EXHIBIT D · THE FAILED DIRECT CROSSING

DENSITY IS NOT RESTRICTION OF MEASURE

Along $j(u) = (p^{iu})_p$ the finite products give positive Euler shadows

$$F_S(u) = \prod_{p \in S} \frac{1 - p^{-1}}{|1 - p^{-1/2 - iu}|^2} = \sum_{q \in \langle S \rangle} \kappa(q) e^{iu \log q}.$$

As length-side measures these do **not** converge to the Weil functional: with $q_n = \frac{n+1}{n}$, $\kappa(q_n) \sim \frac{1}{n}$, so every neighborhood of 0 receives divergent mass; and the support carries arbitrary rational frequencies $\log q$, where the explicit formula has only prime-power lengths $\pm r \log p$.

— POINT OF CONTENTION · READ CAREFULLY

This negative result is not a failure of the program. **It is a correction of the map.** The direct product is too large; it contains all rational-frequency interference terms.

■ EXHIBIT E · THE SHADOW AS RESPONSE

THE STATE MOVES; THE GENERATOR DOES NOT

The same finite object is an autocorrelation. With diagonal flow $U_u|q\rangle = e^{iu \log q}|q\rangle$ and generator $D|q\rangle = (\log q)|q\rangle$,

$$F_{S,\alpha}(u) = \langle \xi_{S,\alpha}, U_u \xi_{S,\alpha} \rangle, \quad R_\alpha(u) = \frac{|\zeta(\alpha + iu)|^2}{\zeta(\alpha)^2} \quad (\alpha > 1).$$

The generator is trivial. **All arithmetic content is in the vector.** As $\alpha \downarrow 1$, $R_\alpha(u) \rightarrow 0$ for $u \neq 0$: the zeta-distribution vacuum forgets itself under every nonzero scale displacement at the pole boundary. At $\alpha = \frac{1}{2}$ the Poisson normalization is precisely the vanishing factor $\zeta_S(1)^{-1}$ against $|\zeta_S(\frac{1}{2} + iu)|^2$ — not a probability limit; the beginning of a renormalization problem.

■ EXHIBIT F · THE LOGARITHMIC DERIVATIVE

THE TRANSFORMATION THAT REACHES THE EXPLICIT FORMULA

The product has rational frequencies; its logarithm has only prime powers. Radial differentiation at the critical exponent isolates

$$\mathcal{P} = \sum_{p, r \geq 1} (\log p) p^{-r/2} (\delta_{r \log p} + \delta_{-r \log p}) = -\partial_\alpha \nu_\alpha|_{\alpha=1/2},$$

the finite-place prime-power component of the explicit formula, equivalently the critical radial derivative of the response's Lévy measure.

Poisson product $\longrightarrow$ logarithm $\longrightarrow$ radial derivative $\longrightarrow$ prime powers

— SAFETY RAIL 3

Positivity of F_S cannot be transferred directly to the Weil functional. Positivity supplies the geometry; the response supplies the state reading; the logarithmic derivative supplies the prime side. **Differentiation need not preserve positivity.**

■ EXHIBIT G · COMPLETION

TATE SUPPLIES THE GAMMA TERM, NOT WEIL POSITIVITY

At the real place, the Gaussian gives the archimedean factor:

$$\int_{\mathbb{R}^\times} e^{-\pi x^2} |x|^s d^\times x = \pi^{-s/2} \Gamma(s/2) = \Gamma_{\mathbb{R}}(s), \quad Z(\Omega, s) = \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

The product formula $|q|_\infty \prod_p |q|_p = 1$ locks the real and finite normalizations together. Unconditional — and yet the completed zeta integral is not the Weil quadratic form. The completed object has three ingredients:

completed Weil distribution = pole terms + gamma term – prime-power term.

■ THE MISSING PROBLEM, NOW PRECISE · BOUNDARY POSITIVITY

Construct the completed, renormalized log-scale distribution W whose finite-place component is the prime-power distribution, whose archimedean component is the gamma contribution, and whose pole terms match the completed explicit formula. Then prove $W(h^{**}h) \geq 0$ for every test function h on log-scale. By Weil's criterion this positivity is equivalent to RH. **Not a final technical step — the locked chamber.**

■ EXHIBIT H · THE GATE

WEIL POSITIVITY AS POSITIVE TYPE ON LOG-SCALE

With $x = e^t$ and $h(t) = e^{t/2} g(e^t)$, the Mellin transform on $(0, \infty)$ becomes the Fourier transform on $\mathbb{R}$ along the critical line: $\widehat{g}(\frac{1}{2} + iu) = \widehat{h}(u)$. The square-root normalization, in its cleanest form. If RH holds, every zero has $1 - \bar{\rho} = \rho$ and the zero side becomes a sum of squares:

$$Q(h) = \sum_{\rho} H(\rho) \overline{H(1 - \bar{\rho})} \xrightarrow{\text{RH}} \sum_{\gamma} |\widehat{h}(\gamma)|^2 \geq 0.$$

If a zero leaves the line, the evaluation is no longer at a real Fourier frequency and the square-norm reading breaks. This is the first-principles reason Weil positivity is RH-shaped.

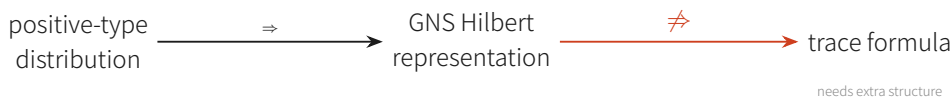


FIG. 5 A GNS VECTOR STATE IS ALREADY A LENGTH REPRESENTATION. IT IS NOT AUTOMATICALLY A TRACE FORMULA — AND HILBERT-PÓLYA WOULD REQUIRE STILL MORE.

Connes's trace-formula program and the Connes–Consani–Marcolli pairing are attempts to build that extra structure. They do not make the gcd-kernel positivity equivalent to Weil positivity.

■ EXHIBIT I · THE BOUNDARY-STATE PROBLEM

ENTERED AS OPEN

THE CORRECTED BOUNDARY QUESTION

Does the family of finite-place vacuum responses admit a renormalized boundary limit — archimedean gamma term and pole corrections included — whose logarithmic derivative is the completed Weil distribution, and whose positivity is **inherited from a genuine state or trace construction**?

A probabilistic avatar of ξ is not automatically Weil positivity (Biane–Pitman–Yor; Nakamura). Probability of the function and positivity of the functional are different doors. Saksman–Webb show the critical line has a rigorous renormalized probability theory — validating the regime, supplying no deterministic Weil-positive state. Connes–Consani work the archimedean bank; the prolate-operator work with Moscovici discretizes the same question from the other side. The finite-place response above is the other bank of the same crossing: trivial scaling generator, nontrivial state, boundary renormalization.

Tomita–Takesaki manufactures canonical time from an algebra and a state. It does not manufacture the Riemann zeros. The conditional statement: if an arithmetic von Neumann algebra and faithful state realized the completed explicit formula, modular theory would supply the canonical generator — and the decisive problem would still be positivity.

THE FOG IS NOT EVERYWHERE. IT HAS ONE GATE.

$$W(h^* * h) \geq 0$$

By Weil, that gate is RH.

THE WHOLE RISK OF THIS MOTION · IT DESERVES TO STAND ALONE

Weil won because the curve's cohomology existed to be decreed. For the number field, no comparable cohomology is entered here, and the record does not pretend otherwise. A decree only relocates difficulty; it does not dissolve it. But the burden changes type — from an infinite audit to a single identity — and the change redirects the search.

■ PART III · PRAYER FOR RELIEF

GRANTING IT REQUIRES BELIEVING NOTHING UNPROVED

Motions do not introduce evidence. They ask the court to change the question being tried. The motion is addressed to the field’s self-understanding of what kind of statement the Hypothesis is.

The relief requested: RH should be approached as a boundary-positivity problem for a completed log-scale distribution. The finite-place gcd kernel supplies a canonical positive geometry at the near boundary. The missing step is not more positivity in general; it is the precise survival, or reconstruction, of positivity after logarithmic differentiation, archimedean completion, and renormalization into Weil’s explicit formula.

A–F
the near bank
CARRIED

G–H
the far condition
FIXED

I
the crossing
OPEN

The zeta zeros stand to arithmetic-at-infinity as the diagonal stood to the Pythagorean world: a failure of cancellation with no address in the standing coordinates. Dedekind did not find the square root of two. He noticed that the absence was itself the shape. The record of the field, ancient and modern alike, says that such absences are resolved in one motion, and that the motion is total: rebuild the field around the impossibility until the impossibility is digested as a type.



The critical line is where arithmetic
should be recentered next.